Analysis results

The simulation data exhibits strong evolutionary pressures where time progression drives high-speed, high-view-range organisms at the cost of reduced physical size and specific dietary requirements.

File

stats.csv

Encoding

utf-8

Columns

12

Total missing values

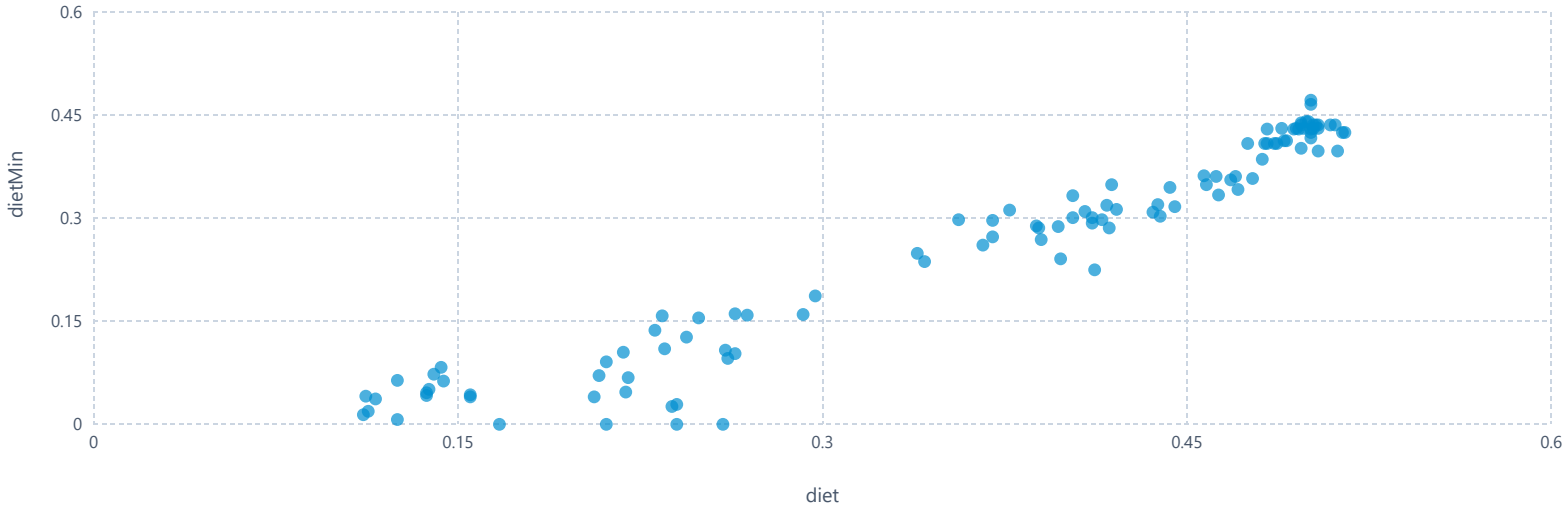
0

ACTIONABLE RECOMMENDATIONS

- Focus on optimizing resource distribution to reduce the death-to-population ratio identified in the creatures vs corpses correlation.
- Introduce environmental variability to test if larger radius sizes can regain a competitive advantage against the current trend of smaller, faster organisms.
- Monitor the extreme decline in plant populations to ensure the simulation doesn't face a resource collapse that would plummet the current creature count.
- Investigate the impact of stabilizing the mutation rate to prevent potential stagnation in biological traits as the system approaches later evolutionary stages.

diet vs dietMin

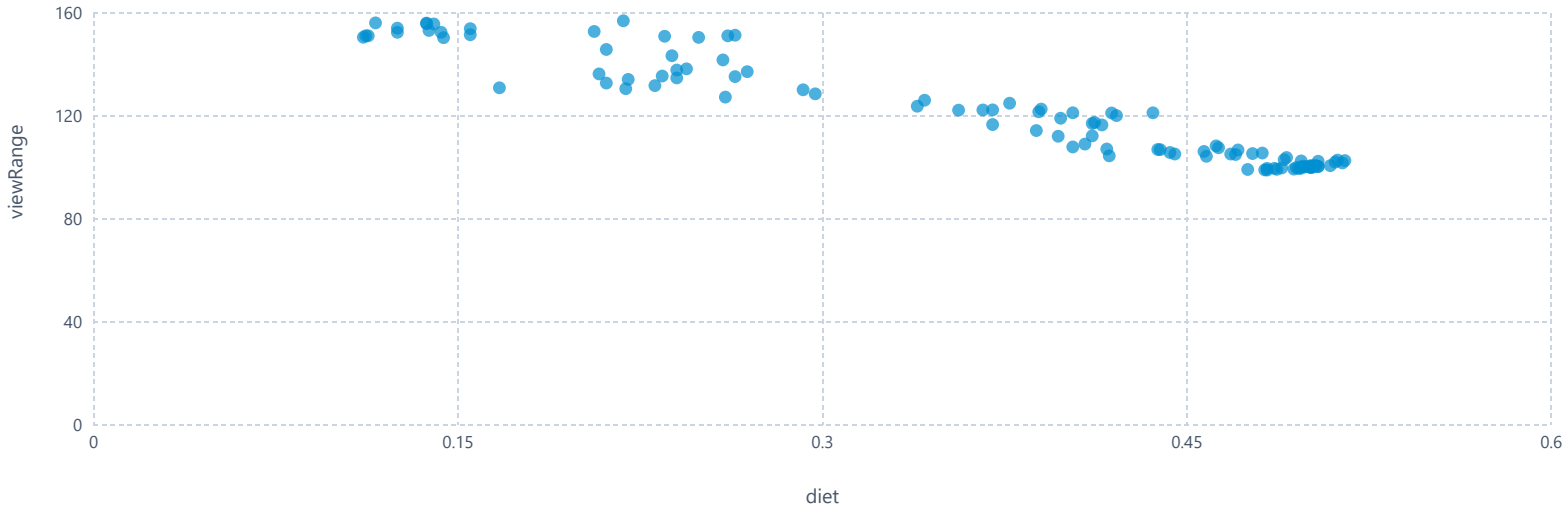
Strong positive



diet vs viewRange

Strong negative

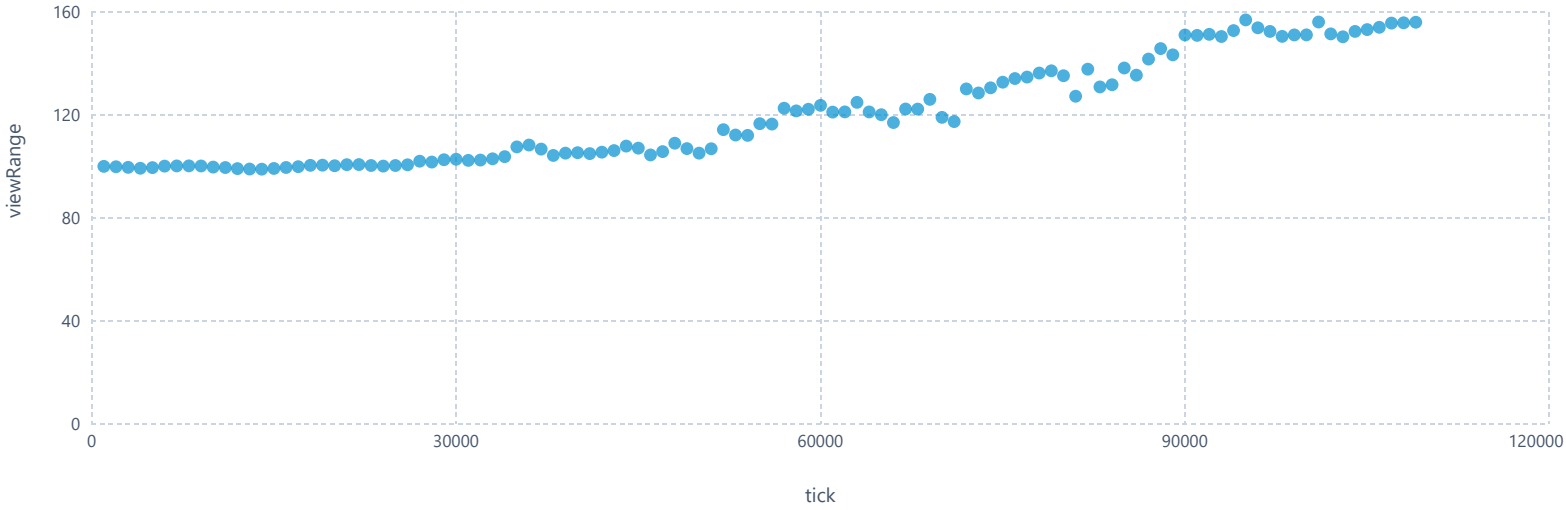
> The near-perfect negative correlation implies that highly specialized or broadly dietary organisms are being outperformed by those with lower diet values as their sensory ranges expand.



tick vs viewRange

Strong positive

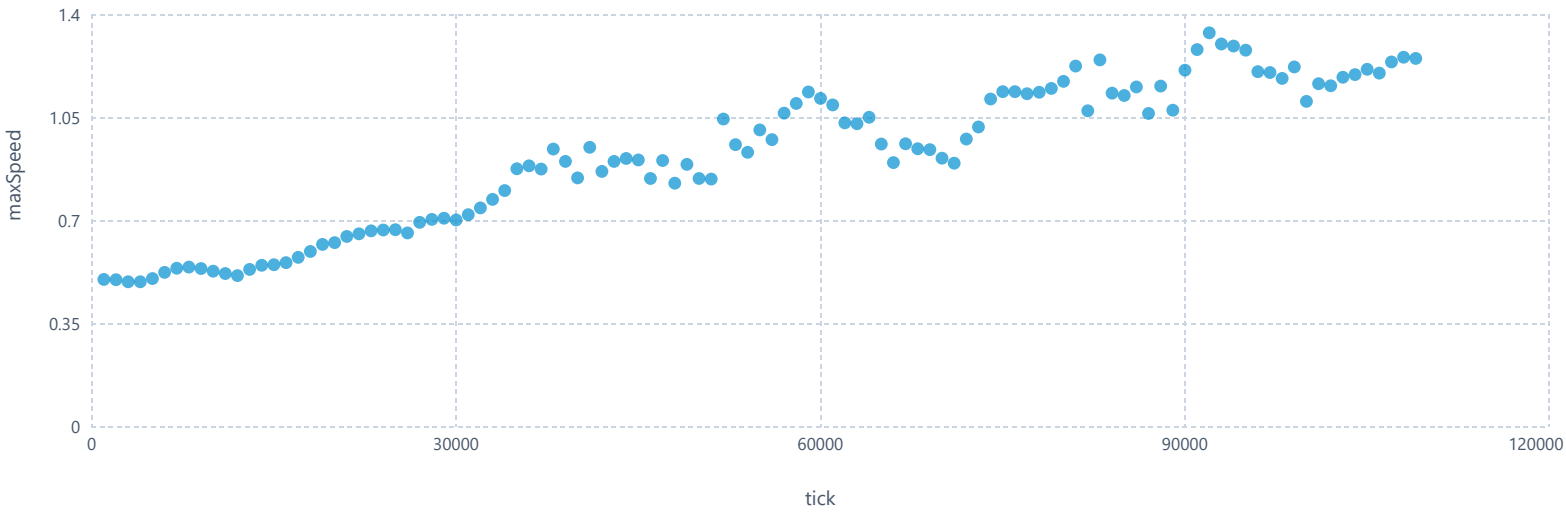
> Increasing view ranges over time indicates that sensory awareness is a critical survival factor, potentially helping organisms optimize foraging or evasion.



tick vs maxSpeed

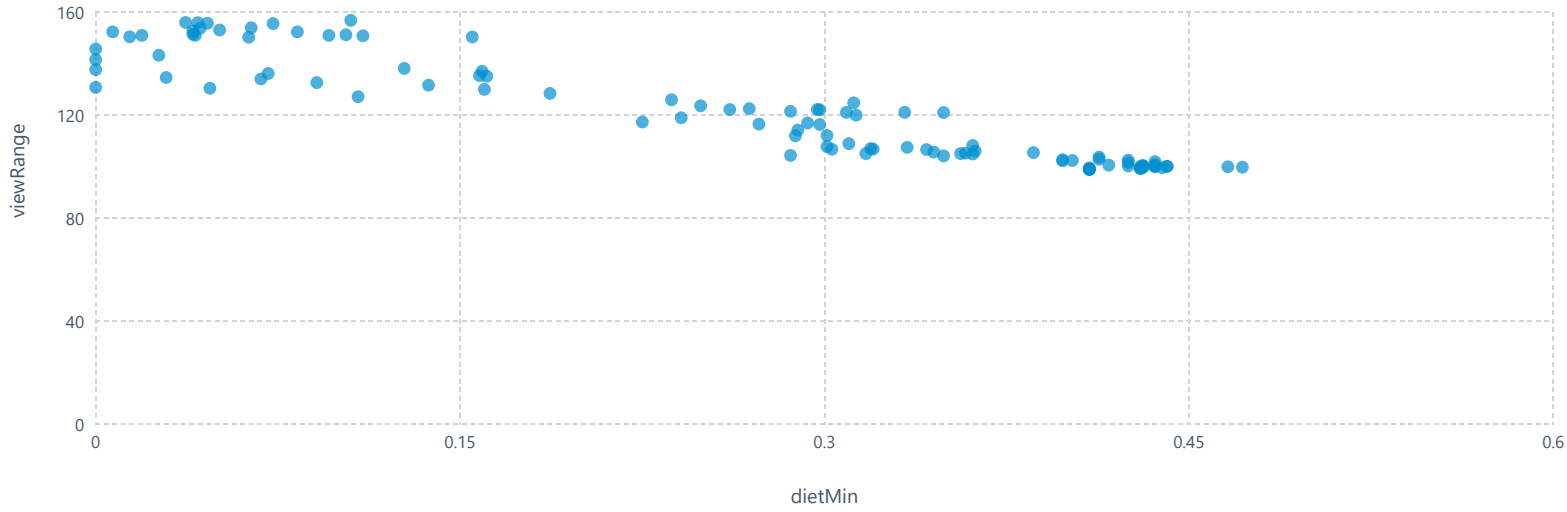
Strong positive

> There is a near-perfect positive correlation between time and speed, suggesting that faster organisms have a distinct survival advantage as the simulation progresses.



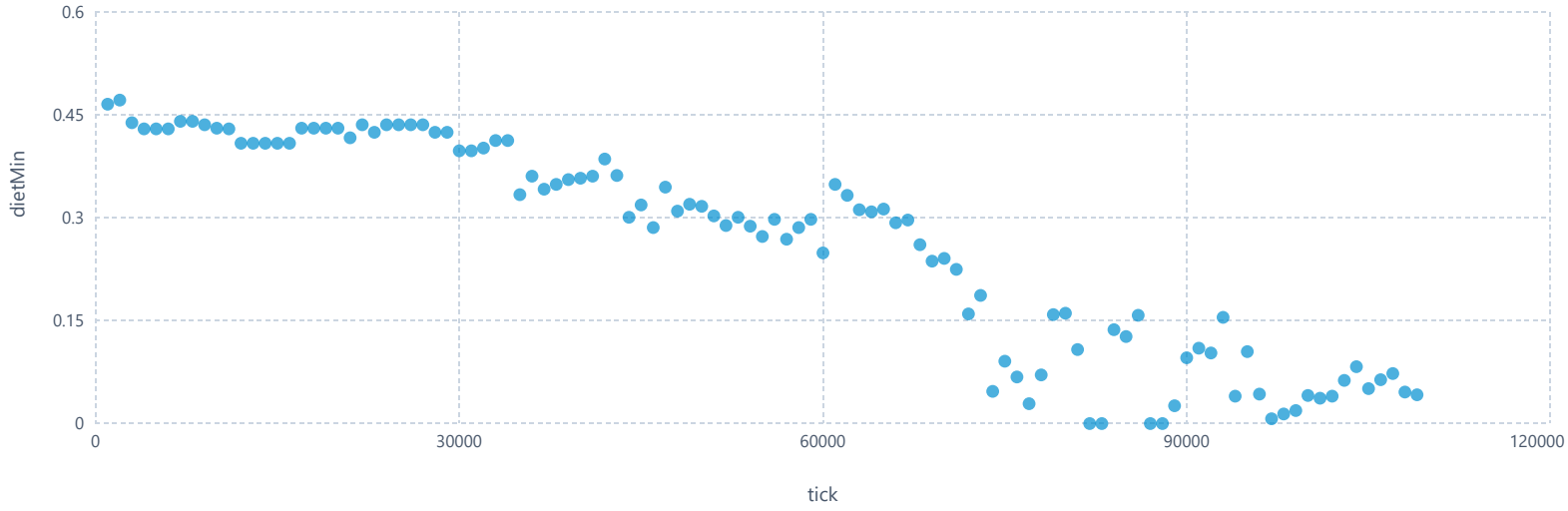
dietMin vs viewRange

Strong negative



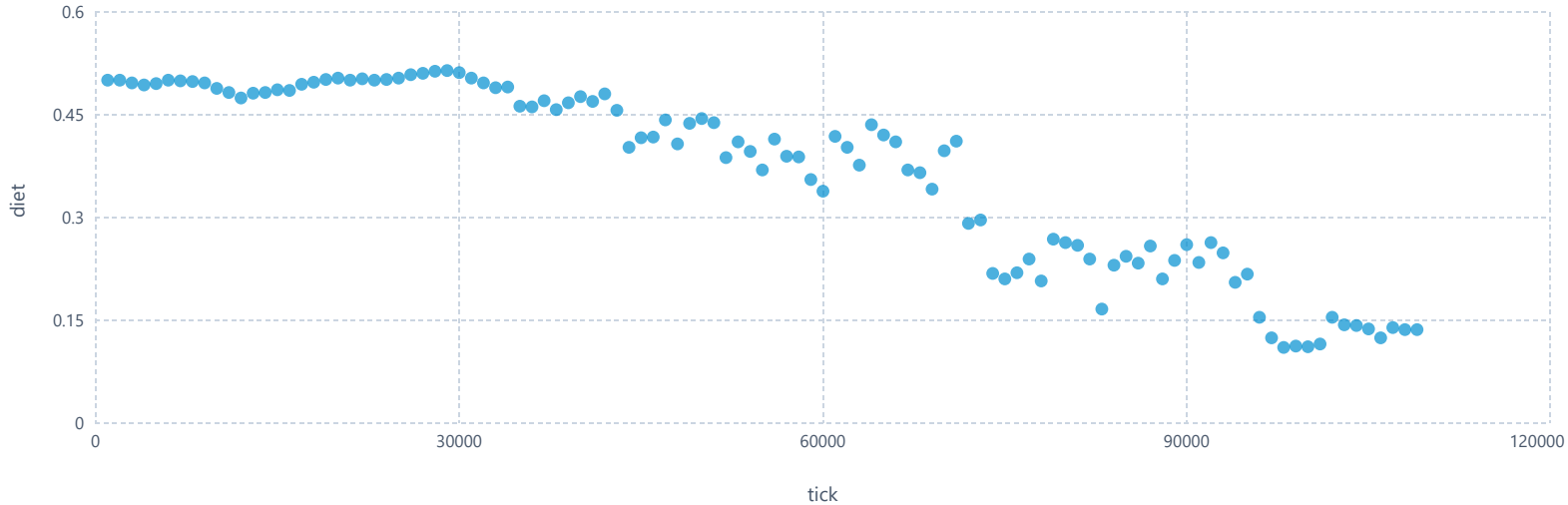
tick vs dietMin

Strong negative



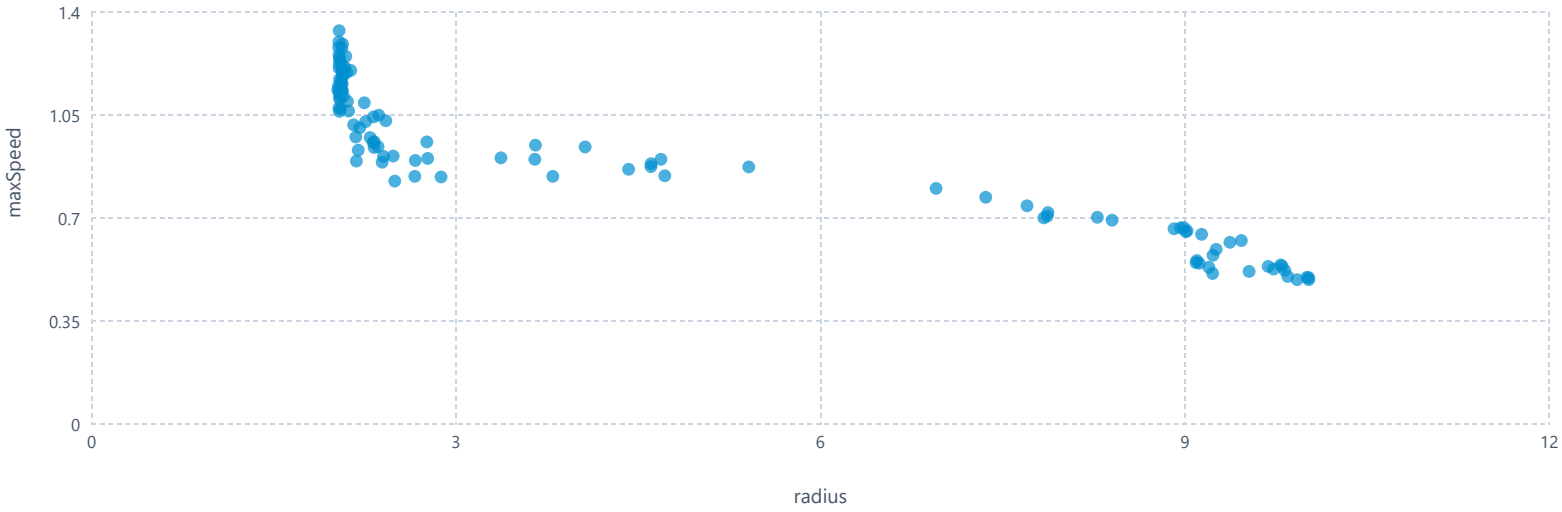
tick vs diet

Strong negative



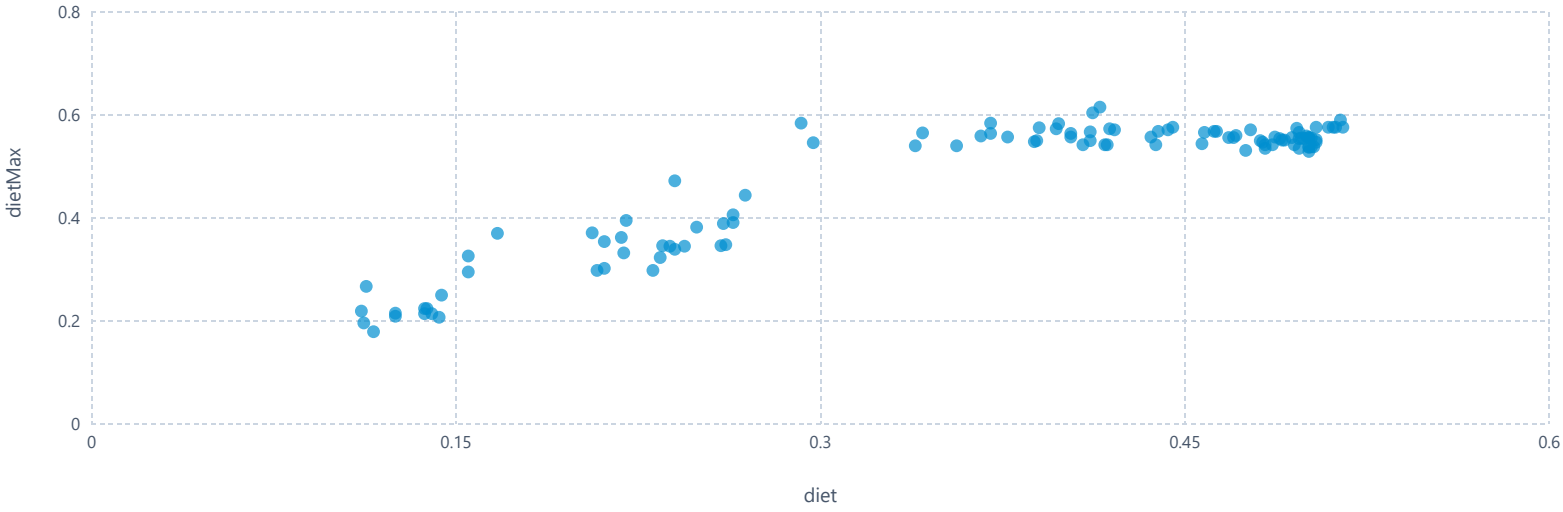
radius vs maxSpeed

Strong negative



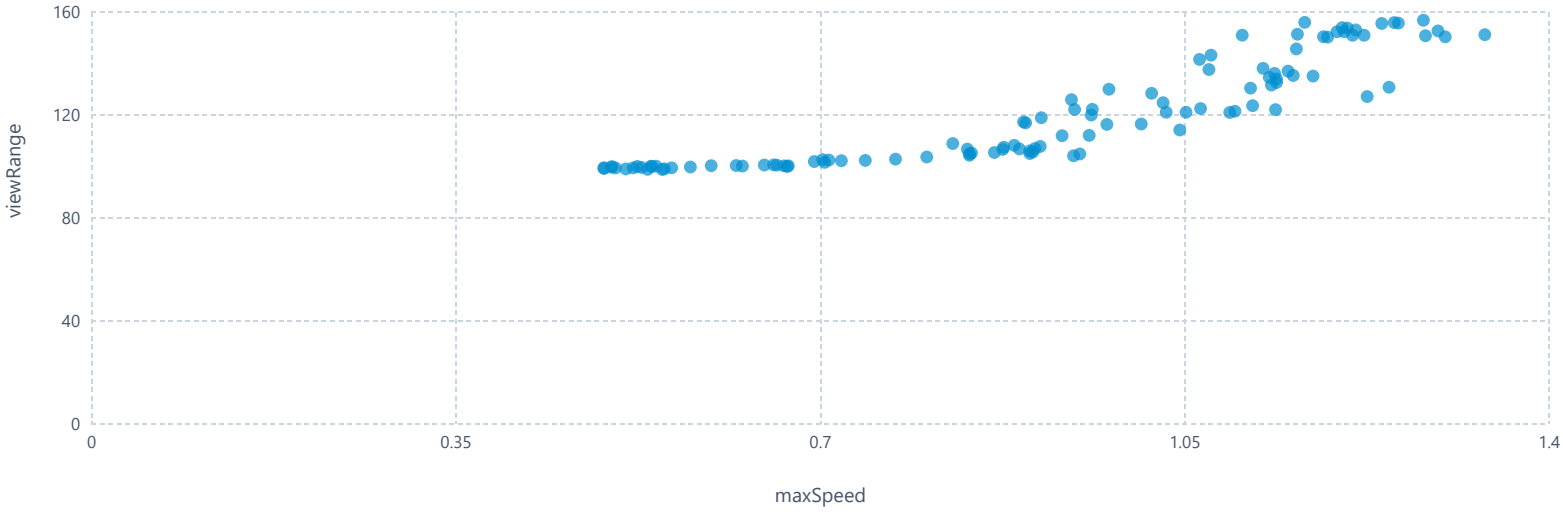
diet vs dietMax

Strong positive



maxSpeed vs viewRange

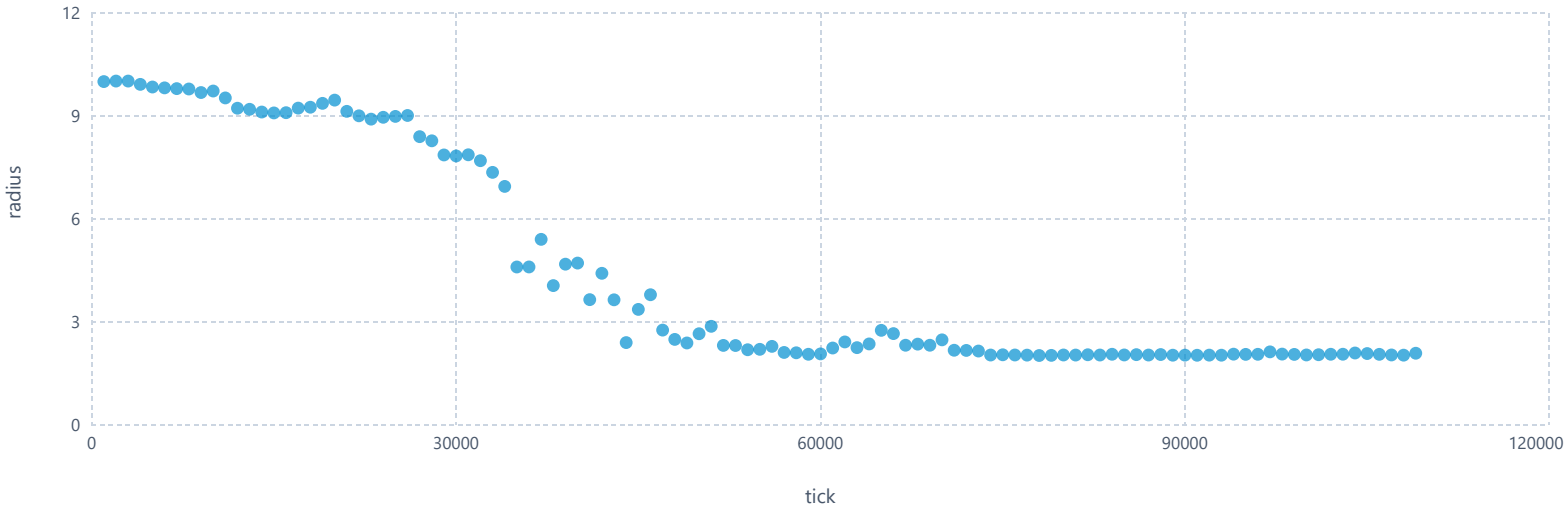
Strong positive



tick vs radius

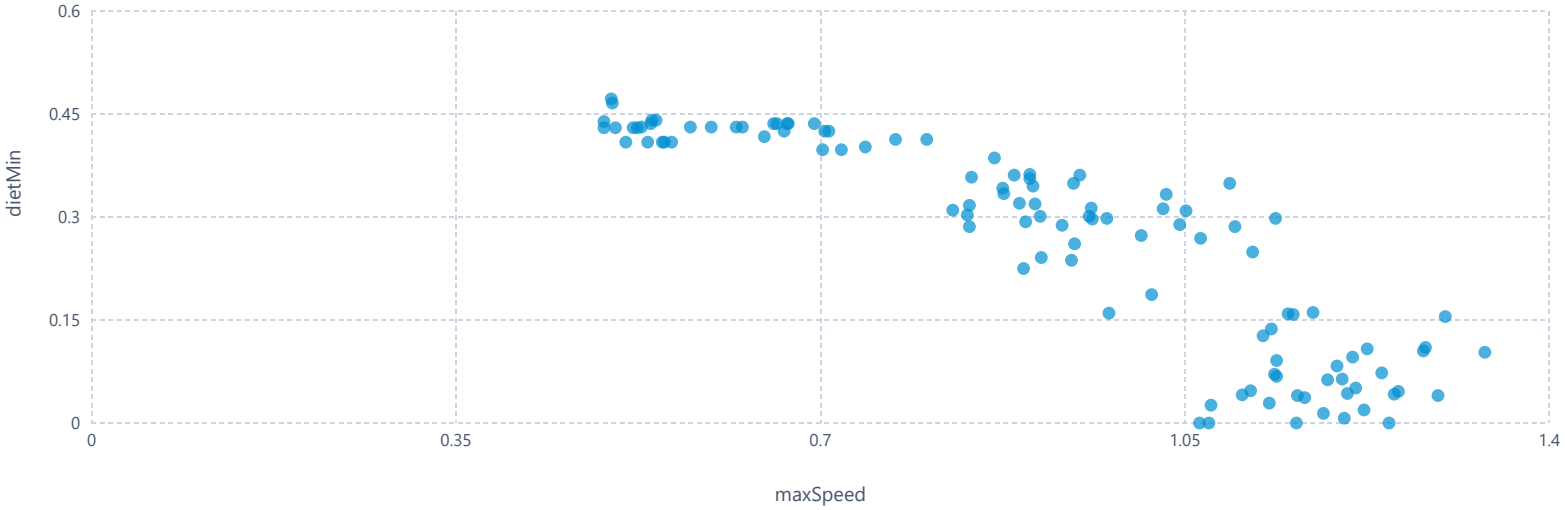
Strong negative

> The strong negative correlation shows a systematic shift towards smaller organisms over time, likely as an efficiency trade-off for the increased speed observed.



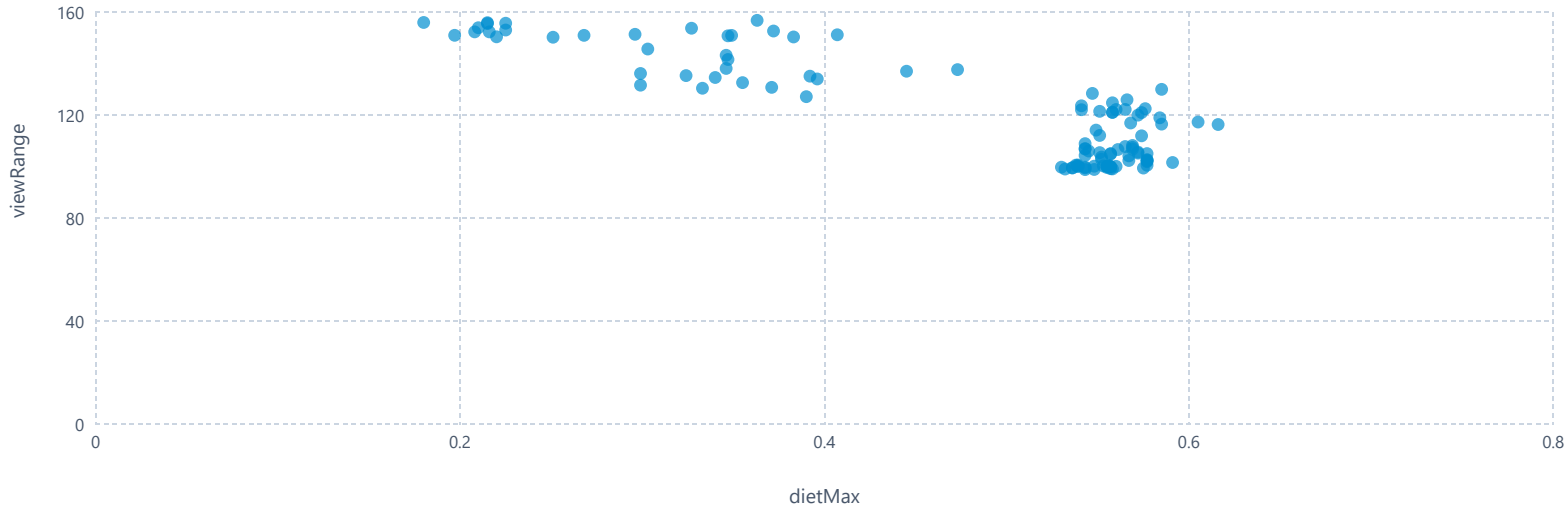
maxSpeed vs dietMin

Strong negative



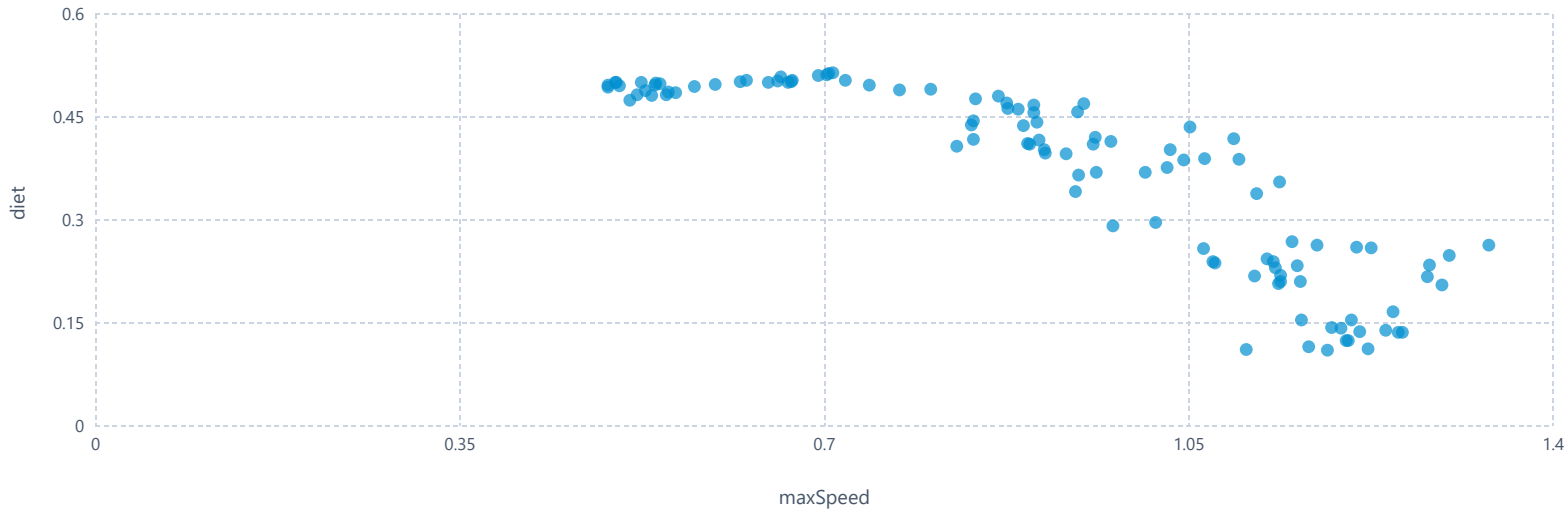
dietMax vs viewRange

Strong negative



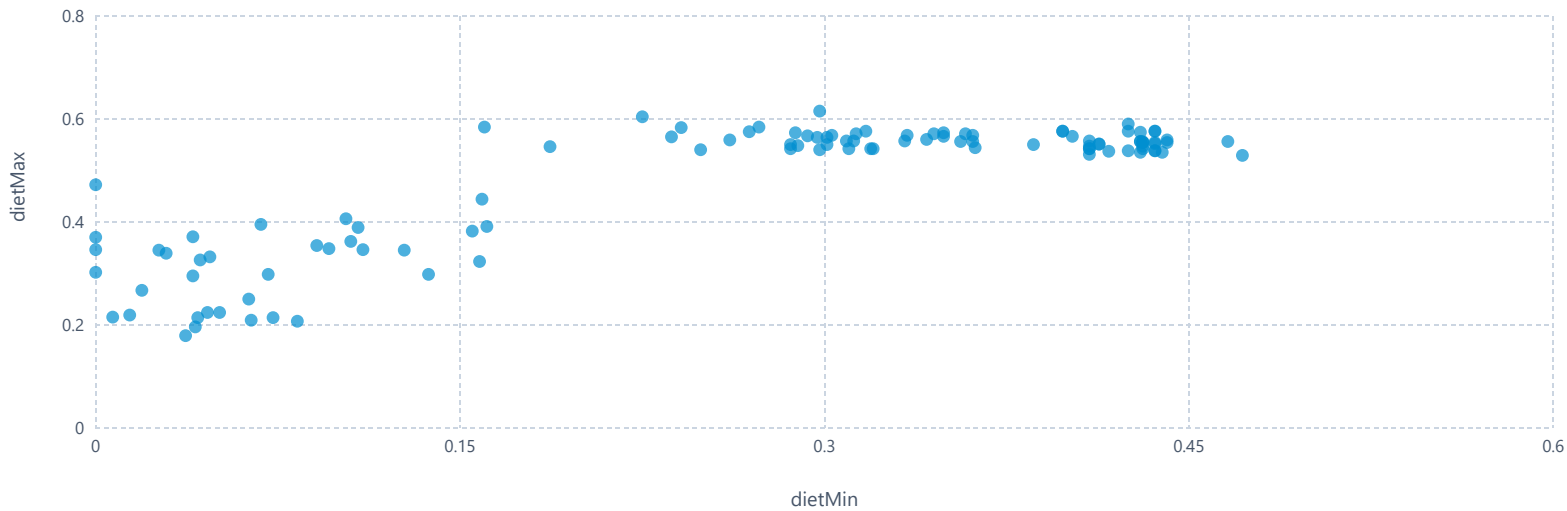
maxSpeed vs diet

Strong negative



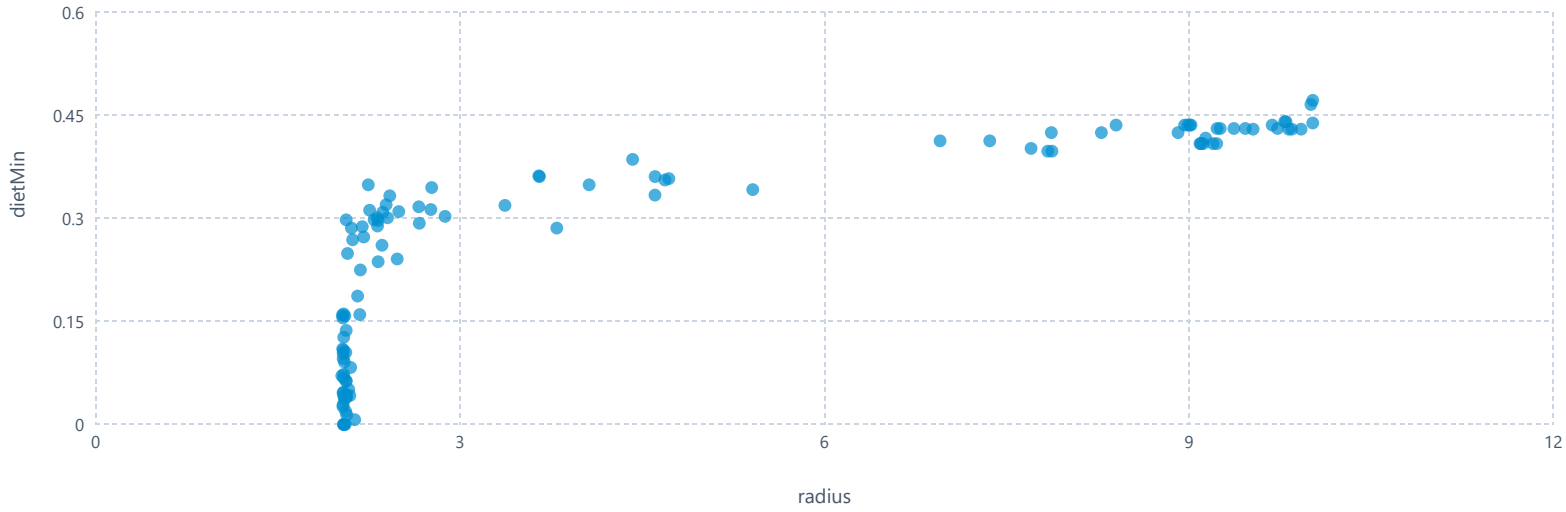
dietMin vs dietMax

Strong positive



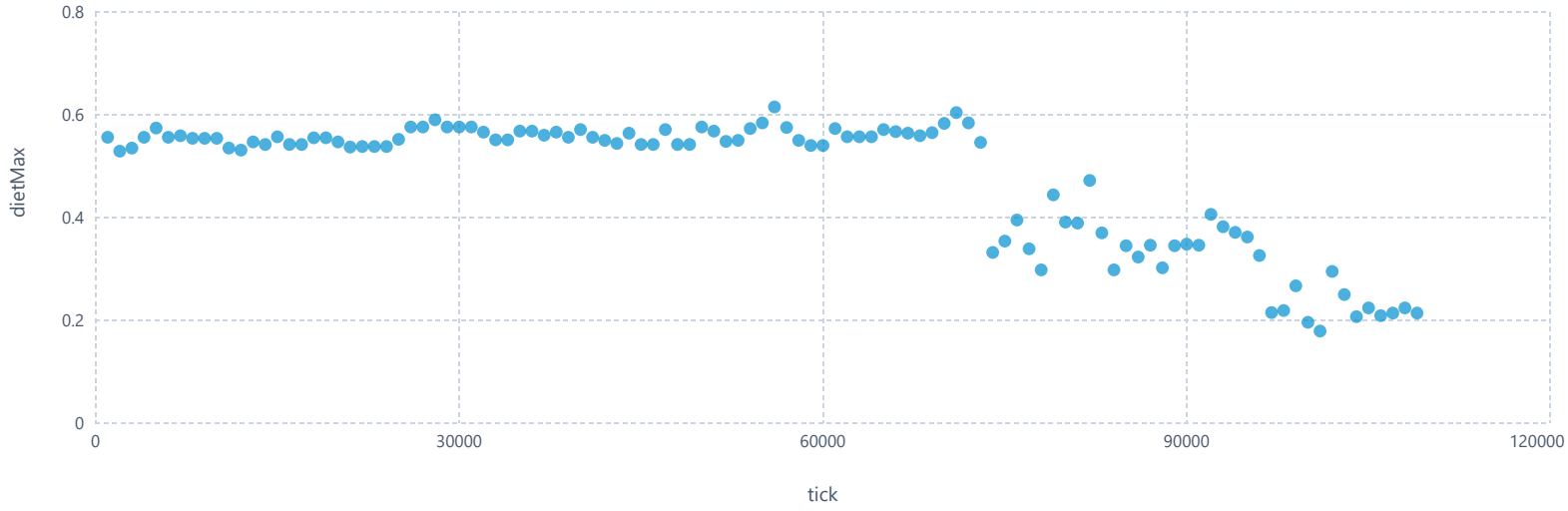
radius vs dietMin

Strong positive



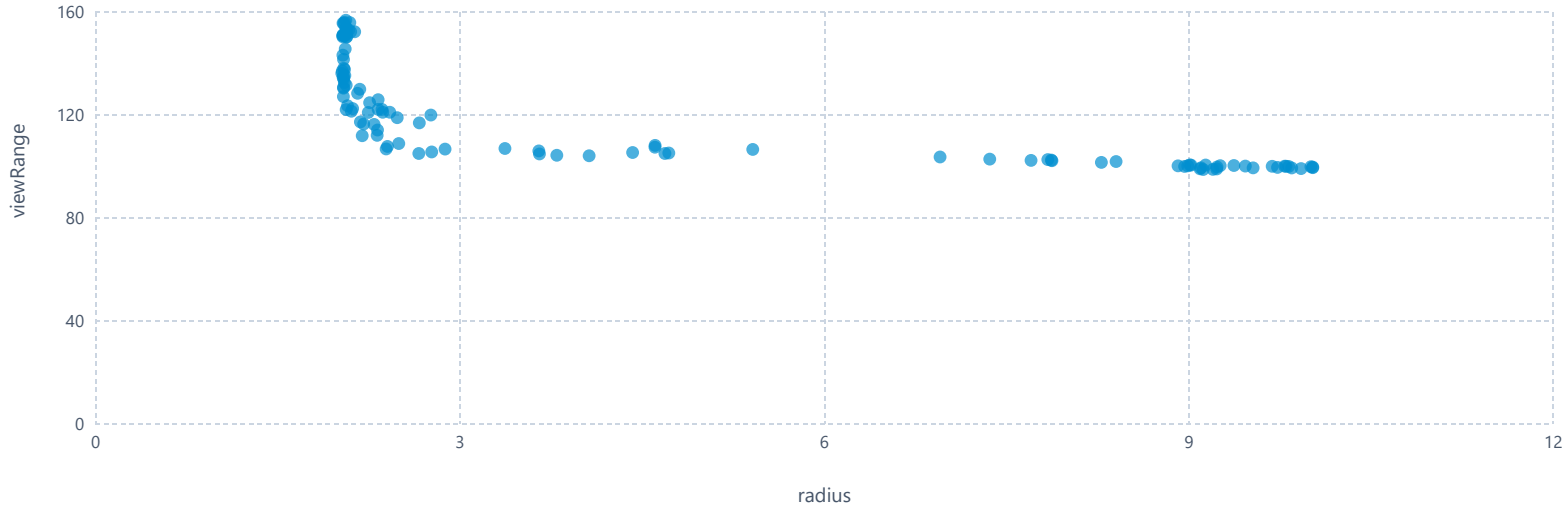
tick vs dietMax

Strong negative



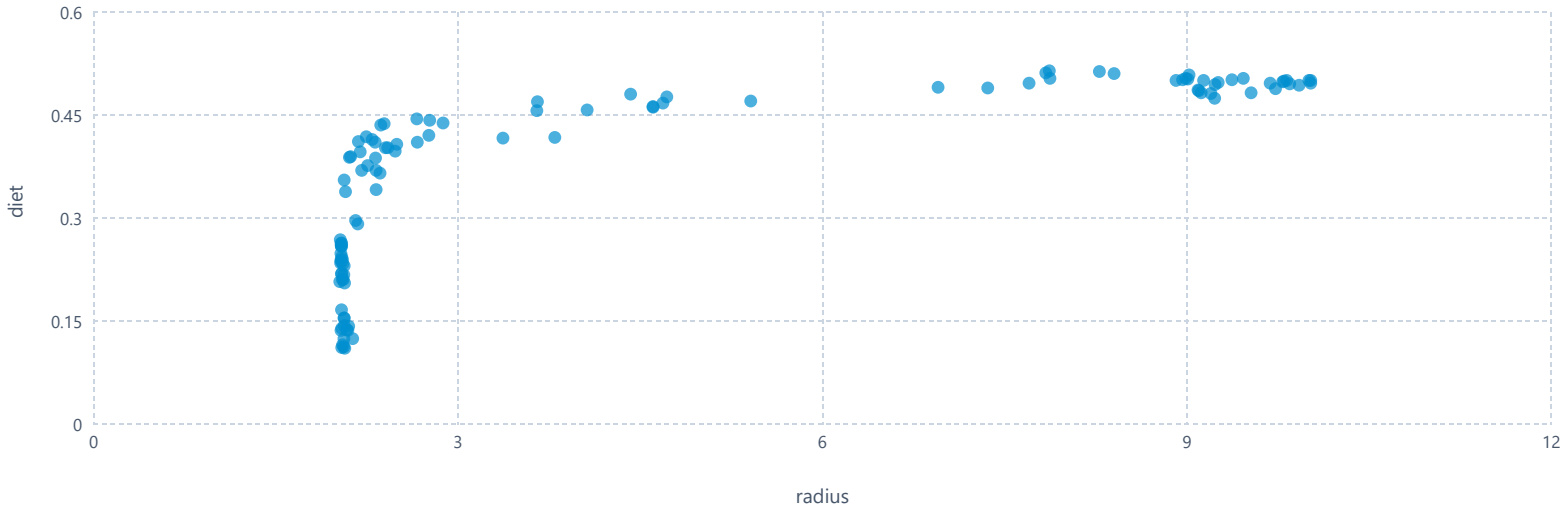
radius vs viewRange

Strong negative



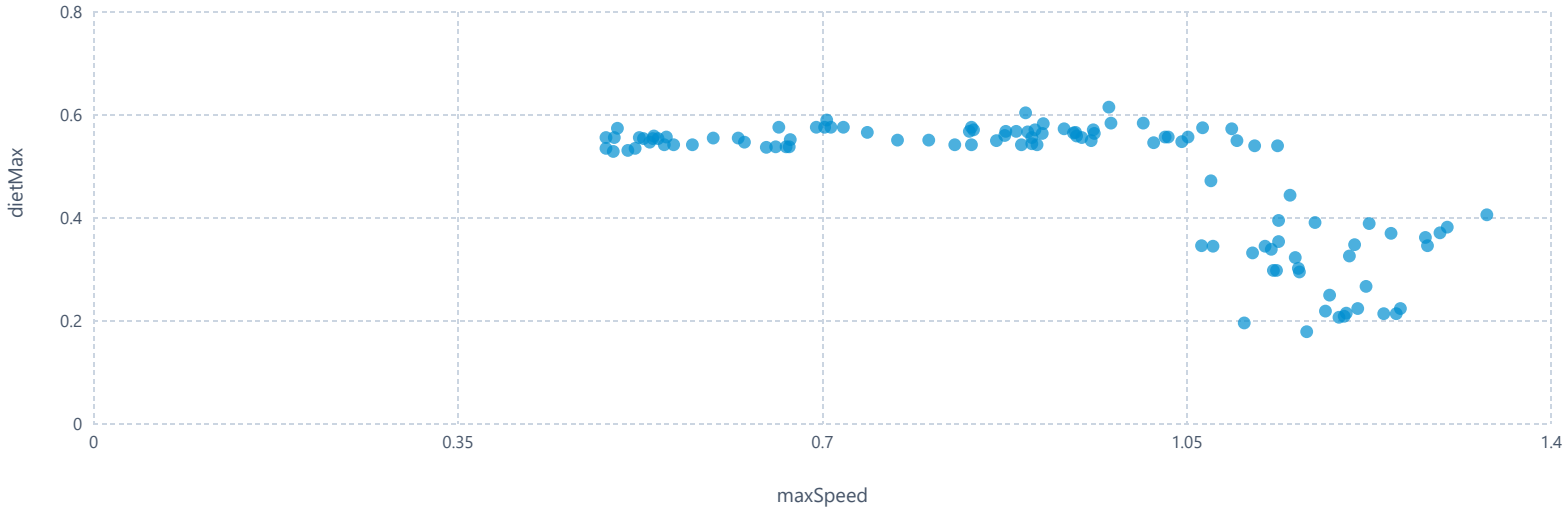
radius vs diet

Strong positive



maxSpeed vs dietMax

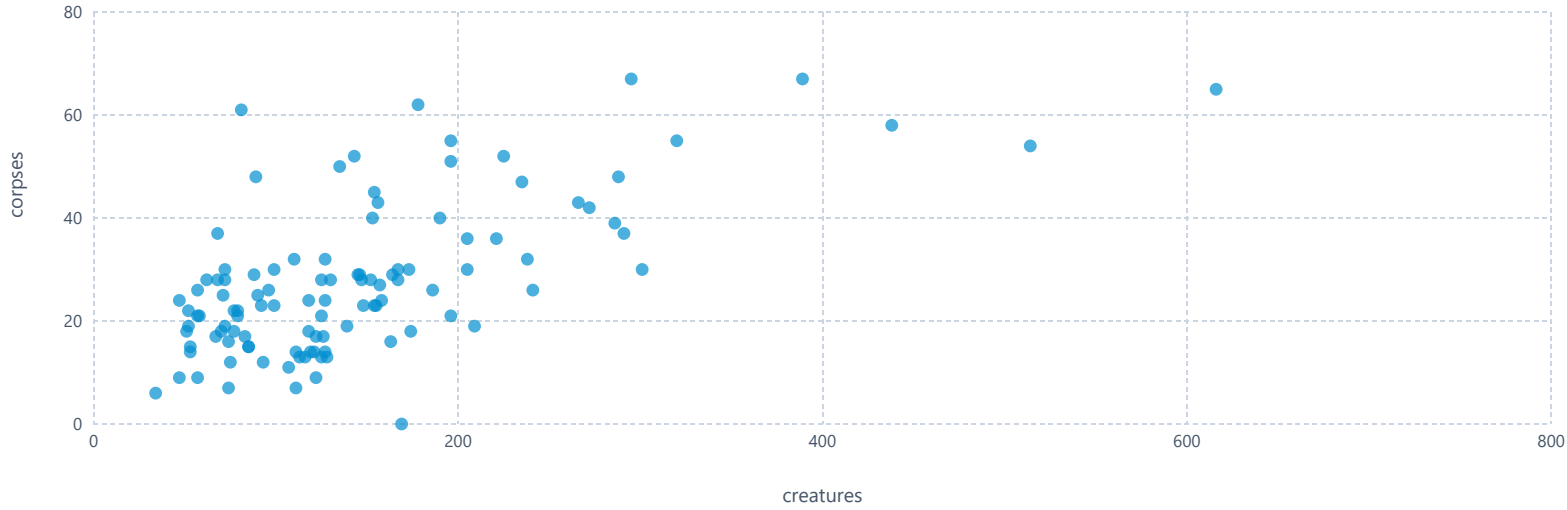
Moderate negative



creatures vs corpses

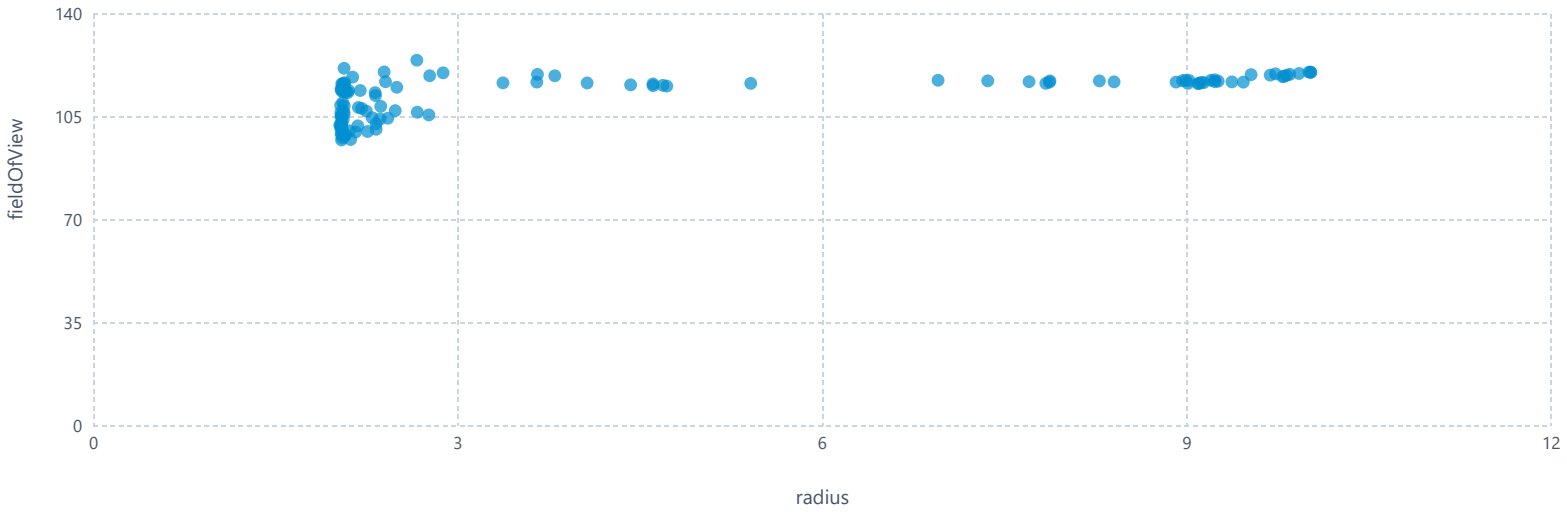
Moderate positive

> The positive correlation confirms that higher creature counts lead to higher mortality, establishing a classic predator-prey or competition dynamic.



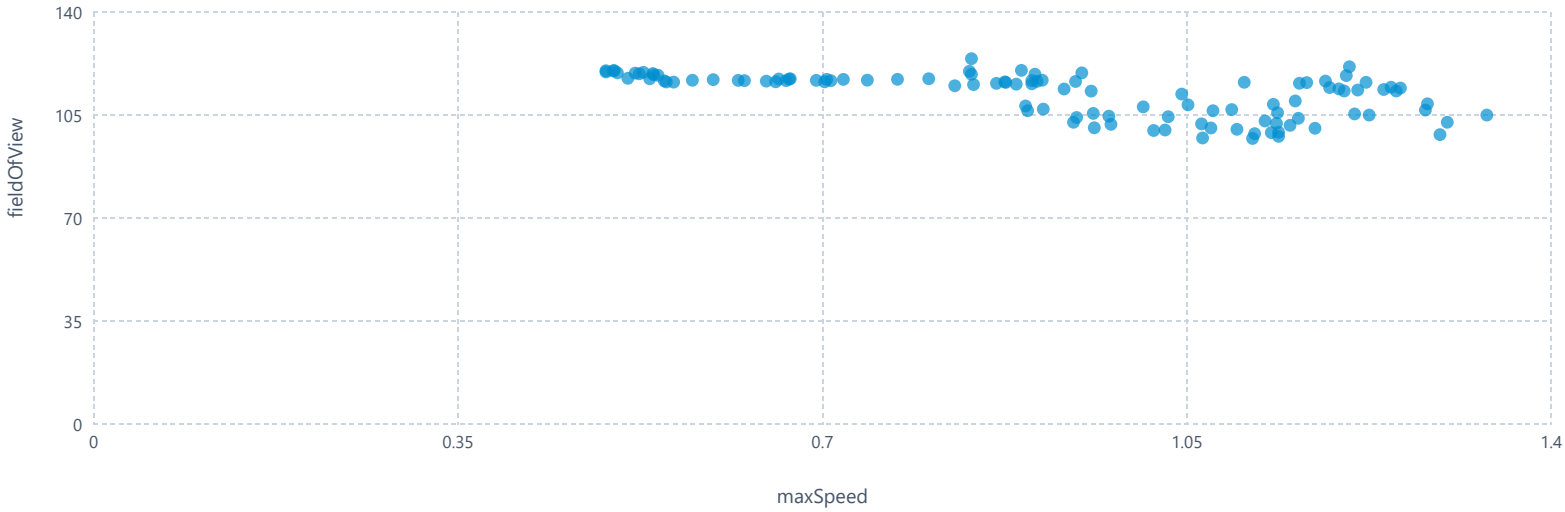
radius vs fieldOfView

Moderate positive



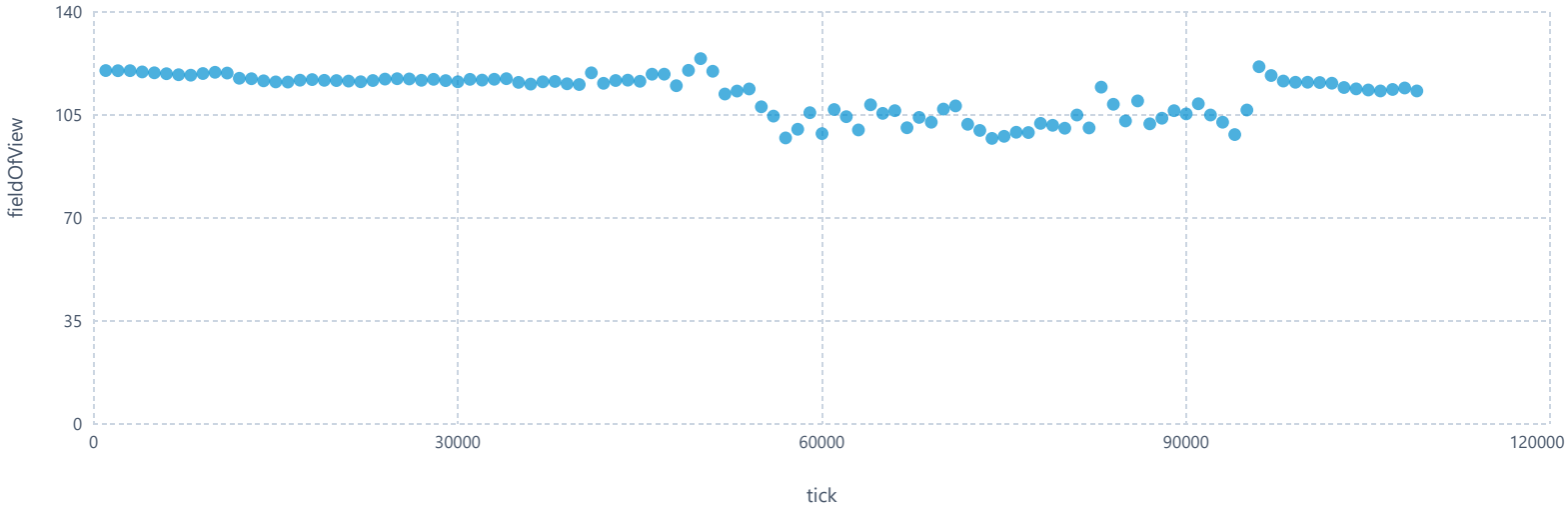
maxSpeed vs fieldOfView

Moderate negative



tick vs fieldOfView

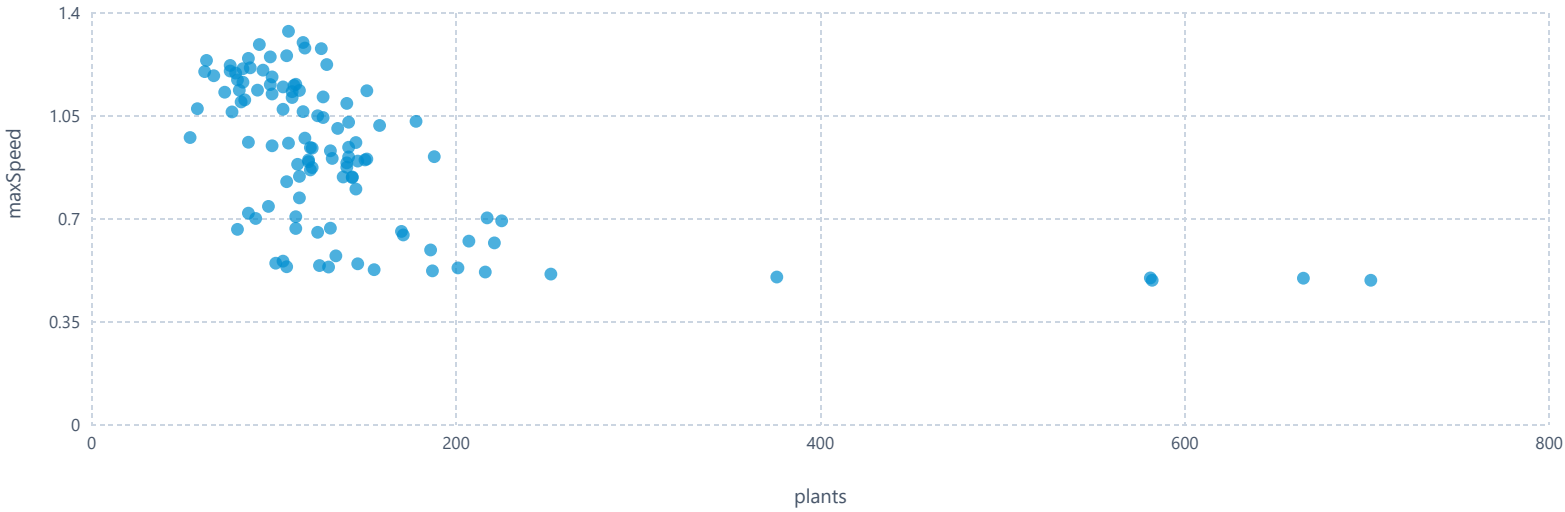
Moderate negative



plants vs maxSpeed

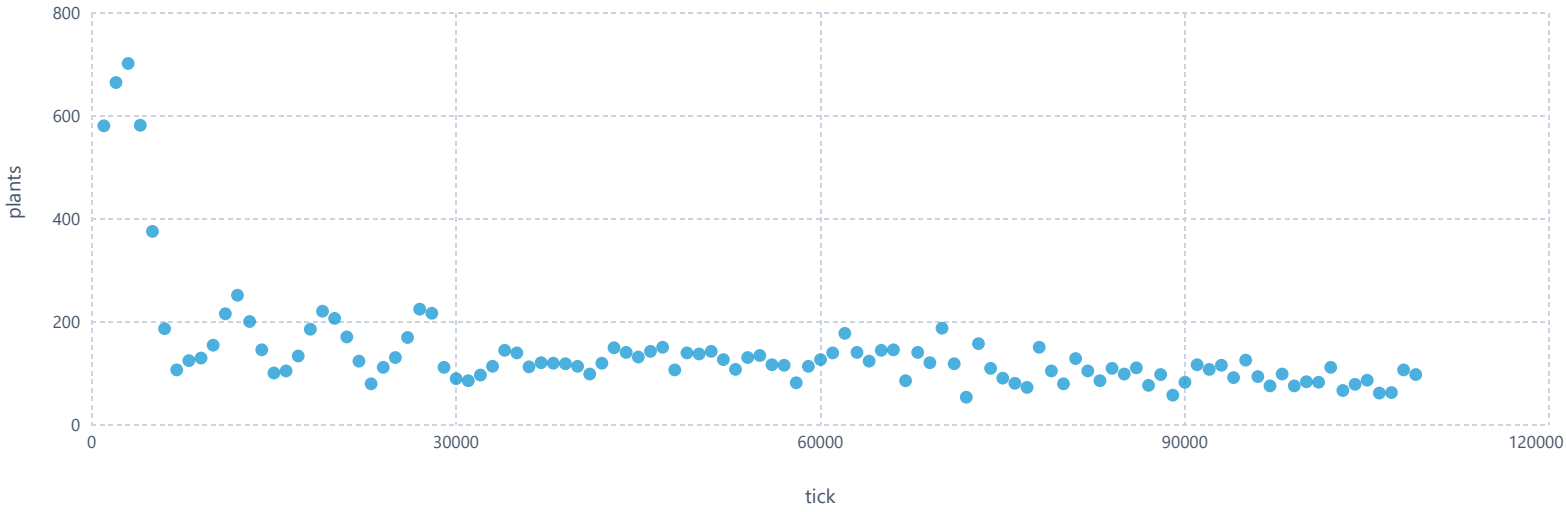
Moderate negative

> The negative correlation suggests that as the population trends toward higher speeds, plant consumption or presence decreases, indicating intense competition for resources.



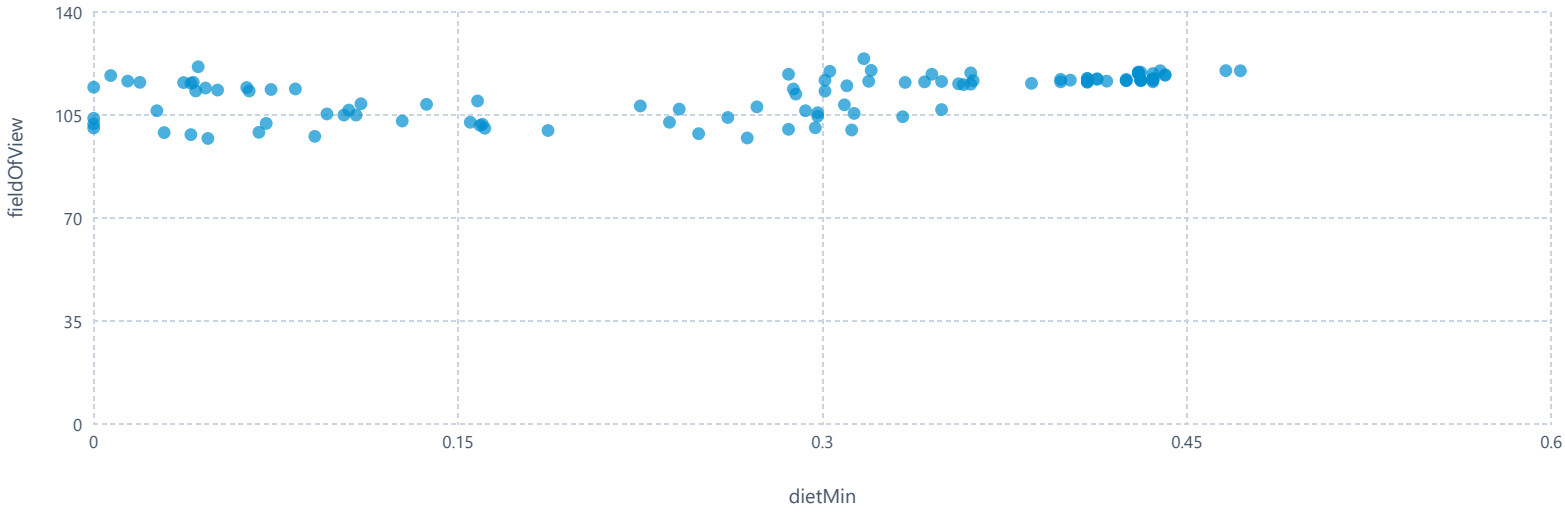
tick vs plants

Moderate negative



dietMin vs fieldOfView

Moderate positive



plants vs radius

Moderate positive

